NASA ARSET - Fundamentals of Remote Sensing for Air Quality

MODULE 1

Air quality impacts both human and environmental health. Satellite remote sensing plays a pivotal role in monitoring and understanding air pollution from local to global scales. Understanding the impacts of poor air quality on human health, the environment and economic activities highlights the need to proactively monitor, study, and mitigate these consequences.

Increased air pollution decreases air quality, which impacts daily activities and long-term health.

Air quality is an assessment of how safe or harmful the air is to people and the environment. It is based on the presence and amount of pollutants in the air.

Air pollution refers to substances in the air that can cause negative impacts to people's health, the environment, and/or the economy.

Poor air quality occurs when large amounts of pollution are present, and good air quality occurs when little pollution is present. Air quality is determined by both the amounts and types of air pollution and describes how healthy or harmful the air is for people and the environment.

Air pollution impacts Increased risk of respiratory illness like asthma, Increased risk of stroke, heart attack, Poor crop growth and health, Productivity loss from related illness

The World Health Organization (WHO) estimates that poor air quality causes seven million premature deaths across the globe each year (as of 2020) (opens in a new tab)

Studies show that higher air pollution leads to reduced crop yields due to reduced photosynthesis, slower growth and increased susceptibility to disease.

Air pollution impacts economies by lowering workforce productivity, and contributing to acid precipitation which damages structures.

Economic growth in the **U.S.** since 1960 has led to increased use of vehicles, higher energy consumption, and a larger population: all of which could lead to worse air quality. Yet, pollutant concentration trends since 1970 have decreased in the US via policies (e.g., Clean Air Act 1963, 1970, 1990) and technological advances (EPA Trends Report).

Remote sensing fills in data gaps between ground-based monitors and enables tracking of the global transport of pollutants.

Direct measurements come from instruments such as ground-based monitors, airborne platforms, and satellite remote sensing. Beyond direct measurement, air quality can also be estimated through laboratory analyses of

collected samples and atmospheric models. Ground-based measurements serve as the reference standard for air quality measurements, including those by satellite remote sensors.

Similar to weather stations measuring temperature and wind, ground-based instruments are used to monitor different air pollutants that determine the air quality.

- Since they are expensive to operate, many low-income countries have few if any in-situ air quality monitors.

- Densely populated areas in high-income countries often have many in-situ air quality monitors.

Measuring air pollution with in-situ monitors is the best way to understand poor air quality where it directly impacts people and the environment.

The expense of operating in-situ monitors means that, on a global scale, the measurement of air quality is intermittent and uneven. Each monitor captures conditions at a single fixed location. Inconsistent and spatially limited data can make it difficult to understand global impacts and find effective solutions

Satellite remote sensing observations can be used to fill in gaps between in-situ air quality monitoring sites in order to help us understand global air quality, its causes and effects, and how it is changing.

Satellite remote sensing can **complement in situ air quality monitors** by:

- Providing a broader spatial data coverage, especially in remote and rural areas

- Tracking air pollutants across long distances (e.g., between continents)

- Identifying the likely sources of air pollution

- Assessing the effectiveness of air pollution control strategies to improve air quality at national and international scales

MODULE 2

For air quality, the layers of the atmosphere closest to the surface are of greatest interest.

The bulk of Earth's atmosphere extends about 100 km above sea level, but some layers are considered to reach thousands of kilometers into space. When studying air quality, the lowest layers of the atmosphere are of greatest interest: the troposphere and the planetary boundary layer. Satellites in space have to look through the other layers of the atmosphere to “see” what is happening where we breathe.

Troposphere: The lowest layer of the atmosphere, extending up to about 10 km altitude. This is where humans, plants, and animals live and breathe.

Planetary Boundary Layer (PBL): The lowest layer of the troposphere. The height of the PBL varies from a few dozen meters to a few kilometers above the Earth's surface, depending on the topography and time of day.

Earth's atmosphere is a mixture of **gases** and suspended solid and liquid particles called **aerosols**.

More than 99.9% of the atmosphere consists of three gases: Nitrogen (N_2), Oxygen (O_2) and Argon (Ar). The negative impacts of poor air quality are caused by the **concentration of the trace gases and aerosols** in the remaining 0.1% of the atmosphere.

Nitrogen (78%)
78%

Oxygen (21%)
21%

Argon (0.9%)
0.9%

Trace gases & aerosols (0.1%)

Trace Gases: All the gases in the atmosphere other than Nitrogen, Oxygen, or Argon.

Aerosols: Microscopic solid or liquid particles suspended in the air.

Air quality studies focus on the small percentage of trace gases and aerosols that impact human and environmental health in the troposphere and planetary boundary layer.

Sources of air pollution can be **anthropogenic** (human-caused) or **natural.** Air pollutants that are directly emitted by a specific source are primary air pollutants. Secondary air pollutants are formed by chemical reactions in the air.

Anthropogenic Sources: Originate from human activity.

Natural Sources: Originate from naturally occurring environmental events and processes.

PRIMARY POLLUTANTS: Directly emitted into the air by human or natural activity.

Transportation
Anthropogenic

Vehicles produce aerosols and trace gases through fossil fuel exhaust, as well as aerosols from non-exhaust sources like brake and tire wear.

Industrial Activity
Anthropogenic

The manufacturing industry creates both aerosol and trace gas air pollution through energy use and byproducts from the materials they process.

Burning of Fuels for Power
Anthropogenic

Power plants produce aerosol and trace gas byproducts that are a source of air pollution.

Agricultural & Land Management
Anthropogenic

In agricultural practices, livestock digestion, fertilizers, and manure are key sources of trace gases. Smoke from fires used for agricultural burning and land management (prescribed burning) is another example of an anthropogenic pollutant.

Residential
Anthropogenic

Residential sources of air pollution include heating such as wood stoves and open fires for cooking, as well as lawn equipment, consumer products, trash burning, and fireworks.

Fires from Natural Events
Natural

Wildfires are primarily considered a natural source when ignited by lightning. However, they can also result from intentional human causes (e.g., arson) or unintentional ones (e.g., discarded cigarettes, campfires).

Dust
Natural

Mineral dust from sources like deserts and arid soil are lofted in the air by high-wind storms.

Pollen
Natural

Pollen from trees and other plants can cause mild to severe respiratory allergies in humans and animals.

Ash from Volcanoes
Natural

Volcanic eruptions may produce ash and other pollutants.

SECONDARY POLLUTANTS: Form in the air due to chemical reactions and physical processes in the atmosphere.

An example of a **secondary pollutant** is smog, which is formed when primary pollutants emitted from vehicles and industry combine and chemically react in the atmosphere.

Air pollution does not stay at its source. Two key factors that are related to its development and dissipation are **transport** and **lifetime**.

Transport: Pollutants can be moved through the environment. For example, wind can disperse pollutants, while stagnant air can cause them to accumulate.

Lifetime: The lifetime of a pollutant refers to how long it remains in the air. The weather and chemical reactions impacts this lifetime. For example, humidity can cause pollutants to stagnate in one location, rain tends to disperse or remove pollutants faster, and chemical reactions can deplete pollutants.

For example, dust from the Sahara Desert can travel over 8,000 kilometers across the Atlantic Ocean to reach the Americas. Pollutant lifetimes also vary widely; nitrogen dioxide (NO_2) lasts only a few hours, while carbon dioxide (CO_2) can persist for centuries.

Aerosols and trace gases are major categories of air pollutants that are observable from space. Understanding these pollutants is essential for developing effective air quality management strategies. Satellite sensors can measure key trace gases important for air quality, as well as many aerosols and their physical characteristics.

AEROSOLS - Particulate Matter

Aerosol pollutants are also commonly called **Particulate Matter (PM)**. They exist in a wide range of sizes and compositions that impact health in different ways.

Aerosols, commonly referred to as particulate matter (PM), vary in shape, size, and chemical makeup. For air quality applications, aerosols are typically classified by the size of the particles. PM10 refers to particles that are 10 microns or less in diameter and PM2.5 refers to particles are 2.5 microns or less in diameter.

Particulate Matter ≤ 10 mm (PM_{10}): Aerosol particles 10 microns in diameter and smaller. This is also sometimes called **coarse PM.** At this size, they can irritate your nose and throat and enter your lungs.

Particulate Matter ≤ 2.5 mm ($PM_{2.5}$): Aerosol particles 2.5 microns in diameter and smaller. This is also sometimes called **fine PM**. These particles can pass from your lungs into your bloodstream, impacting your heart, arteries, and brain.

Volcanic ash contains **coarse** (PM_{10}) and **fine** particles ($PM_{2.5}$), with larger particles settling quickly while smaller ash can remain airborne and travel long distances.

Mineral dust is dominated by **coarse** particles (PM_{10}) lifted from dry soils and transported regionally or across continents.

Sea Spray aerosols are mainly **coarse** particles (PM_{10}) formed by breaking ocean waves that eject salt droplets into the atmosphere.

Smoke from wildfires and agricultural burning consists primarily of **fine** particles ($PM_{2.5}$) that can remain suspended and spread widely.

Vehicle exhaust produces **fine** and **ultrafine** particles, typically smaller than 1 μm, formed during fuel combustion in engines.

Smog contains fine particles ($PM_{2.5}$) from atmospheric chemical reactions involving nitrogen oxides and volatile organic compounds.

TRACE GASES

Nitrogen Dioxide (NO_2) is usually emitted by burning fossil fuels, especially in vehicles and power plants. It irritates your airways, exacerbating conditions like asthma. It commonly reacts with other gases to form secondary aerosols that create smog.

Carbon Monoxide (CO) is emitted by burning fuels, such as wood or fossil fuels, especially when there is inadequate ventilation. High levels of CO can exacerbate heart conditions by reducing the amount of oxygen in your blood.

Sulfur Dioxide (SO_2) is emitted by burning fuels with a higher sulfur content, like coal. SO_2 is also emitted during volcanic eruptions. In addition to causing negative respiratory symptoms, SO_2 can contribute to acid precipitation, which harms plants and ecosystems.

Ozone (O_3) forms in the lower atmosphere and planetary boundary layer as a secondary pollutant from reactions between NO_2 and other organic trace gases, known as **Volatile Organic Compounds (VOCs)**, in the presence of sunlight. Exposure to ozone causes respiratory issues, is harmful to plants and contributes to smog.

Although there are other trace gases which can impact air quality, NO_2, CO, SO_2, and O_3 are among the most commonly monitored and are also observable with remote sensing. There are other important atmospheric trace gases, such as the greenhouse gases of carbon dioxide (CO_2) and methane (CH_4) that have significant climate impacts.

Air Quality is quantified by measuring the presence and concentrations of air pollutants and comparing these with established standards.

There are three main components for evaluating and communicating useful information about air quality and the impacts of air pollution on human and environmental health.

Measure/Scientific Analysis

Scientists use measurements of aerosols and trace gases at or near ground level to determine concentrations of pollutants in the air.

- **Scientific Analysis**: Scientific studies analyze measurements of pollutant concentrations with human and environmental health data to assess impacts

Governments and non-governmental organizations may define different thresholds for acceptable air quality. There are many different procedures for establishing and enforcing air quality standards for the context and population they are serving.

Air Quality Indexes (AQIs) are guidelines for comparing air quality conditions and determining health risk. The AQI is usually based on the defined standards or guidelines in place for air pollutants in a country. An overall risk is represented by the AQI by combining multiple standards into a single output.

Basic Facts about Air Quality Indexes:

- **Simplify Complicated Information**: AQIs translate complicated information on concentrations of different air pollutants into a single number.

- **Not a Measurement**: AQI is not a direct measure of pollutants.

- **No Universal AQI Scale**: Different countries and organizations use different AQIs though many use similar pollutants to calculate AQI.

The pollutants $PM_{2.5}$, PM_{10}, NO_2, O_3, CO, and SO_2 are the most common indicators used to assess air quality and appear in most air quality standards around the world.

There are several areas in the US where ozone AQI is Moderate or greater. These areas of higher ozone AQI are likely due to emissions from URBAN sources.

The high PM2.5 AQI areas are most associated with the WILDFIRE AND URBAN sources, with high values in the north US but also notable Moderate AQI in high population density corridors.

There are few areas with Moderate or higher PM10 AQI, and sources tend to be from various BLOWING DUST AND INDUSTRY locations.

High ozone usually stems from vehicle emissions reacting with sunlight in urban areas. High PM10 often comes from dust or industrial activities like mining, while high PM2.5 on this date results from Canadian wildfires or local urban sources.

By measuring how components in the air interact with radiation at different optical wavelengths, satellites can detect the presence and movement of pollutants over entire regions. This makes it possible to track air quality on regional, national, and global scales, providing insights that can support public health, policy decisions, and everyday awareness.

Optical Wavelengths: The ultraviolet (UV), visible (VIS), and near-infrared (NIR) parts of the electromagnetic spectrum.

The VIIRS sensor (opens in a new tab) was designed to continue and improve upon measurements initiated by its predecessors. VIIRS sensors are operating on the NASA-NOAA partnership Suomi NPP satellite and two NOAA satellites in the Joint Polar Satellite System (JPSS) series. VIIRS-derived data products are used to study Earth's land, atmosphere, cryosphere, and oceans.

- Data Availability: Starting 2011
- Orbit: Polar-Orbiting **LEO**
- Energy Source: **Passive**
- Spectral Resolution: **Multispectral** (22 bands)
- Spectral Range: Visible and IR (410 to 12010 nm)
- Spatial Resolution: 0.375 to 0.750 km (depending on band)
- Temporal Resolution: 1 day
- Swath Width: 3000 km
- Satellites: SNPP (2011-2026), NOAA-20 (2017), NOAA-21 (2021)

MODIS (opens in a new tab) is a key instrument operating on NASA's polar-orbiting Terra and Aqua satellites for more than 20 years, with morning (Terra) and afternoon (Aqua) overpass times to capture measurements at different times of day. MODIS data products are relevant to air quality (particularly aerosol) and are also used to measure other geophysical parameters across the atmosphere, land, cryosphere, and ocean. Scientists use MODIS data to improve our understanding of global climate, vegetation, land use, oceans, clouds, fires, and other hazards. Note: The Terra and Aqua satellites are near the end of their mission lifetime and could be decommissioned in 2026.

- Data Availability: Starting 2001
- Orbit: Polar-Orbiting **LEO**
- Energy Source: **Passive**
- Spectral Resolution: **Multispectral** (36 bands)
- Spectral Range: Visible and IR (400 to 14400 nm)
- Spatial Resolution: 0.250 to 1.0 km (depending on band)
- Temporal Resolution: 1 -2 days depending on latitude
- Swath Width: 2330 km
- Satellites: Terra (2001), Aqua (2004)

ABI
(opens in a new tab)
is the primary instrument aboard NOAA's GOES-R series of satellites for imaging Earth's weather, oceans, atmosphere, and environment. At any given time, there are two operating satellites in the "GOES-East" and "GOES-West" geostationary positions, providing overlapping coverage of the Americas. These are supplemented by an international constellation of instruments with similar capabilities, creating a global geostationary observing network.

- Data Availability: Starting 2016
- Orbit: **GEO**
- Energy Source: **Passive**
- Spectral Resolution: **Multispectral** (16 bands)
- Spectral Range: Visible to IR (470 nm to 13300 nm)
- Spatial Resolution: 0.5 to 2 km (depending on band)
- Temporal Resolution: 30 s to 10 min depending on mode and region
- Field of Regard: Full disk
- Temporal Resolution: 10 minutes for full disk scans, faster for sub-regions.
- Satellites: GOES-16 (launched 2016) through GOES-19 (launched 2024)

TEMPO
(opens in a new tab)
is NASA's first mission designed to monitor gaseous atmospheric pollutants at hourly temporal resolution during daylight. TEMPO joins a constellation of similar instruments over East Asia (the Geostationary Environmental Monitoring Spectrometer (GEMS) instrument, launched in 2020) and Europe (the Sentinel-4 instrument, launched in July 2025).

- Data Availability: Starting 2023
- Orbit: **GEO**
- Energy Source: **Passive**
- Spectral Resolution: **Hyperspectral** (2048 bands)
- Spectral Range: UV to Visible (290 to 490 nm and 540 to 740 nm)
- Spatial Resolution: 4.75 x 2.0 km
- Temporal Resolution: 1 hr (10 min for special operations) during daylight
- Field of Regard: Greater North America
- Satellites: Intelsat-40e (launched 2023)

CALIPSO was a joint NASA and CNES (French space agency) mission which provided new insights into the role that clouds and atmospheric aerosols play in regulating Earth's weather, climate, and air quality. The
CALIOP instrument
(opens in a new tab)
used an active remote sensor, a lidar, which allowed it to gather information about the vertical distribution of clouds and aerosols. The mission came to an end in 2023, but the historical data are still available, and a new ESA mission, EarthCARE has been making spaceborne lidar observations since 2024.

- Data Availability: Starting 2006-2023
- Orbit: Polar-Orbiting **LEO**

- Energy Source: **Active**
- Spectral Resolution: **Dual-band** (532 nm, 1064 nm)
- Spectral Range: Vis to NIR (1 band each)
- Spatial Resolution: 0.1 km footprint (0.3 to 0.6 km vertical)
- Temporal Resolution: 16 days
- Swath Width: 0.1 km
- Satellite: Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observation (CALIPSO), launched 2006

OMI
(opens in a new tab)
is a contribution of the Netherlands's Agency for Aerospace Programs (NIVR) in collaboration with the Finnish Meteorological Institute (FMI) to the NASA Aura mission. It monitors key air quality components from urban to regional scales. Note: The Aura mission is near the end of its mission lifetime and could be decommissioned in 2026.

- Data Availability: Starting 2004
- Orbit: Polar-Orbiting **LEO**
- Energy Source: **Passive**
- Spectral Resolution: **Hyperspectral** (1560 bands)
- Spectral Range: UV to Vis (270 to 500 nm)
- Spatial Resolution: 13 x 24 km
- Temporal Resolution: 1 day
- Swath Width: 2600 km
- Satellite: Aura (launched 2004)

A European Space Agency mission, TROPOMI
(opens in a new tab)
builds upon the legacy of NASA's Ozone Monitoring Instrument (OMI) mission (launched in 2004) of monitoring the composition of the Earth's atmosphere. Its finer spatial resolution and greater spectral range compared to OMI give it enhanced capabilities for monitoring gaseous pollutants in the atmosphere, which we will discuss later in the module.

- Data Availability: Starting 2017
- Orbit: Polar-orbiting **LEO**
- Energy Source: **Passive**
- Spectral Resolution: **Hyperspectral** (2600 bands)
- Spectral Range: UV to NIR (270 to 2385 nm)
- Spatial Resolution: 7.0 x 3.5 km
- Temporal Resolution: 1 day
- Swath Width: 2600 km
- Satellite: Sentinel-5 Precursor (2017)

The OMPS instrument suite was designed to extend and improve upon long-term atmospheric ozone measurements initiated by heritage sensors like TOMS and SBUV/2. Operating as a joint effort between NASA and NOAA, OMPS sensors fly aboard the Suomi NPP satellite and the Joint Polar Satellite System (JPSS) series. OMPS-derived data products are primarily used to monitor global total column ozone, vertical ozone concentration profiles, sulfur dioxide (SO_2), and atmospheric aerosols to track stratospheric health and air quality. The OMPS Nadir Mapper sensor characteristics are listed here as it is the most comparable to OMI.

- Data Availability: Starting 2011
- Orbit: Polar-Orbiting **LEO**
- Energy Source: **Passive**
- Spectral Resolution: **Hyperspectral**/Spectrometer (~0.4 nm to 1.0 nm)
- Spectral Range: Ultraviolet (300 to 380 nm)
- Spatial Resolution: 50 km × 50 km (SNPP), 12.5 km × 17.5 km (NOAA-20), 10 km × 8.75 km (NOAA-21)
- Temporal Resolution: 1 day (global daily coverage)
- Swath Width: 2800 km
- Satellites: SNPP (2011–2026), NOAA-20 (2017–present), NOAA-21 (2022–present)

To have many observations per day at the same location, an instrument on a geostationary satellite is needed, such as the GOES ABI, but global measurements require a polar-orbiting satellite and wide swath width such as the VIIRS instrument on the NOAA satellite series. For the height of a feature such as a smoke or ash plume, an active sensor is required such as the CALIOP lidar.

Aerosols interact with radiation in the optical parts of the electromagnetic spectrum. Their presence and optical properties can be measured with multispectral sensors.

When measured with in-situ sensors, aerosols are typically classified by size, and their presence is quantified using size-based measurements like **$PM_{2.5}$** and **PM_{10}**. Aerosols can also be classified by their chemical composition, where their presence is quantified by a type-specific mass concentration.

Satellite remote sensors used for air quality observations cannot measure these properties of aerosols directly. Instead, they measure how aerosols interact with light in the near infrared (IR), visible (VIS), and ultraviolet (UV) parts of the electromagnetic spectrum. These optical properties can be used within a **retrieval algorithm** to infer the total amount of aerosols present in the atmosphere and distinguish the most prevalent aerosol types (e.g., smoke, dust, or urban pollution).

A retrieval algorithm is the set of steps and calculations that use a sensor's raw measurements to derive or 'retrieve' a geophysical parameter.

Individual aerosol particles are too small to see, though high concentrations, or **plumes**, made up of these particles (such as smoke or dust) can be seen by our eyes because they block visible light. For example, a hazy sky is a visible indication that aerosol particles are scattering or reflecting light compared to a clear, bright, blue sky.

Satellite sensors use visible light interacting with aerosols to measure their presence in the atmospheric column, from surface to space. This optical property is quantified using a geophysical parameter called **Aerosol Optical Depth (AOD)**.
The AOD product overlaid on True Color RGB imagery presented in this image depicts large amounts of aerosols in central and north Africa, as indicated by the red coloring at the end of the white arrows.

AOD TERMS:

“A” for Aerosol
All types of aerosols (e.g., dust, smoke) attenuate light, and therefore all contribute to the AOD.

"O" for Optical
Optical wavelengths, i.e., those near the visible portion of the electromagnetic spectrum, are used to measure how much sunlight is scattered and absorbed by aerosols.

"D" for Depth
The term ‘depth’ doesn’t mean physical thickness, but instead how much aerosols affect sunlight from the surface all the way up through the atmosphere. It’s like asking ‘how opaque is the atmosphere because of aerosols scattering and absorbing light?'

Characteristics of AOD:

1. 1**AOD is dimensionless**, with values of 0 or higher. An AOD of 0 indicates no aerosols are present. Low values (less than 0.4) indicate a relatively small amount of aerosols in the atmosphere, while higher values (0.5 or greater) indicate larger amounts of aerosols are present. While AOD has no maximum value, the highest values typically observed are around 5.

2. 2**AOD can be measured at any optical wavelength**. However, if a wavelength is not specified, it is usually assumed that this refers to AOD derived using the mid-visible wavelength range (500-550nm), which well represents the total aerosols.

3. 3**Highly reflective surfaces and atmospheric conditions can interfere with AOD retrievals,** for example surfaces like snow or water, and the presence of thick clouds or smoke.

Coarse Aerosols Dominant:
For large aerosols, the outgoing light wavelengths (short and long) are scattered or absorbed more evenly.

Fine Aerosols Dominant:
For small aerosols, the outgoing shorter blue wavelength light is scattered or absorbed more than the longer red wavelength light.

Angstrom Exponent (AE): AE is a parameter describing the change in AOD across a wavelength range (often calculated using two pairs of wavelengths in the visible, typically 440nm and 870nm).

Basic Characteristics of Angstrom Exponent (AE)

1. 1Values of 1.5 or higher indicate an atmospheric column dominated by fine aerosols like smoke and urban pollution (e.g., diesel emissions, power plants, or domestic wood burning).

2. 2Values of 0.5 or lower indicate coarse aerosols like mineral dust and sea salt are more prevalent.

- **Distinguishing Aerosol Types: Absorbing vs. Non-Absorbing**
- Certain types of aerosols, like smoke and dust, tend to absorb more sunlight, i.e., they are more absorbing at optical wavelengths; while others like volcanic ash, sea salt, and many industrial and combustion emissions tend to be more scattering at the same wavelengths. Measurements of radiation in the UV part of the spectrum provide information about these effects.
- **Absorbing Aerosol Index (AI)**
- To distinguish between absorbing and non-absorbing aerosols an Absorbing Aerosol Index ("Aerosol Index" or AI) product is derived from the measured UV reflectances at different wavelengths. These measurements in the UV range can be made over bright surfaces (like snow, sun glint areas, or above clouds) where AOD retrievals using visible wavelengths are difficult.
- **The image shown here continues the same scene**. Above, AOD and AE had missing values over clouds and bright desert. These same areas have a range of positive AI values in yellow to red, which indicate the presence of absorbing aerosols.

Absorbing Aerosol Index (AI): AI is a dimensionless value indicating the presence of absorbing aerosols, such as smoke and dust, based on measurements in the UV. It is most often referred to as 'Aerosol Index' (AI), but can also be known as the UV Aerosol Index (UVAI).

Characteristics of Aerosol Index (AI):

1. 1Positive values denote presence of absorbing aerosols (e.g., smoke, dust), but not the total concentration or mass of aerosols.

2. 2AI increases when absorbing aerosols are more concentrated, but it can also rise when these aerosols are lofted higher, making the UV signal easier for the satellite to detect.

3. 3Values near 0 (or negative) denote little to no absorbing aerosols, but AOD could still be high due to scattering aerosols from urban pollution, volcanic emissions, and sea salt.

AI works best for detecting aerosols in large smoke plumes and dust storms high in the atmosphere, like the smoke from biomass burning shown here over central Africa. It's less useful for tracking everyday urban pollution, which tends to stay closer to the ground and contains a mix of different particle types (scattering & absorbing).

Remote sensing of trace gases uses the gases' unique absorption at specific wavelengths. Dips and spikes in the total observed spectra reveal where gases like water vapor, carbon dioxide, and nitrogen oxides absorb

energy, each contributing distinct spectral signatures. Satellite instruments use hyperspectral measurements and iterative retrieval algorithms, combining radiative transfer models with observed data, to separate the overlapping spectral signals and estimate the trace gas concentrations.

Radiative Transfer Model (RTM): An RTM is a computer-based simulation that describes how energy from the Sun travels through the atmosphere and interacts with gases, clouds, aerosols, land, and water before being measured by a satellite sensor. The RTM is used within a **trace gas retrieval algorithm** to allow the scientist to work backward from what the satellite sees to determine what gases are actually present in the atmosphere.

Why Remote Sensing for NO_2 is Useful

Satellite observations of tropospheric column NO_2 generally correlate well with surface-based NO_2 measurements. This correlation is due to most NO_2 sources being at the surface, and its short atmospheric lifetime, leading to large concentrations near its source regions. NO_2 can be a precursor to ozone formation near the surface, and thus, remote sensing of NO_2 provides a valuable tool to anticipate poor air quality due to ozone.

Passive remote sensing observations of trace gases and aerosols typically observe the entire atmospheric column.

he only statement supported by the true color imagery and AOD is that some type of aerosol is obscuring the view of the surface in central India (location 4).
In southern Afghanistan (location 1), the true color imagery appears free of clouds and gives an appearance of a desert surface and a relatively small area of high AOD exists. While northern India and eastern Pakistan (location 2) have features that look like clouds or fog in the true color imagery, the AOD product indicates high amounts of aerosols are present. Clouds or fog would likely block detection of aerosols in the area. In northwestern India (location 3) the surface can be clearly seen in the true color image, but the AOD product indicates a moderate to high presence of aerosol as well. In central India (location 4), the AOD product also indicates high amounts of aerosols.

Data Processing

Data processing takes the raw spectral signals from remote sensors and generates successive levels of data products.

Level 1: Satellite Measured Spectral Radiance

Level 0 raw spectral data are georeferenced and calibrated to produce Level 1 data for the instrument's spectral channels.

True color images are made by combining the red, green, and blue spectral channels of sensors when those are available.

To produce Level 2 products, a retrieval algorithm is applied to the Level 1 data.

Retrieval algorithms are based on physical knowledge, including the spectral signatures of gases and aerosols, the detected light's path through the atmosphere, and other assumptions such as reflectivity of Earth's surface. Different retrieval algorithms may result in different values for the same geophysical parameter because they use different mathematical approaches, inputs, or assumptions.
Quality flags that accompany the data product communicate information about uncertainties in the retrieval algorithm's output.

Level 2: Geophysical Parameter (Swath)

Level 2 products contain geophysical parameters that are derived using a retrieval algorithm.
Level 2 data are georeferenced with the satellite overpass, so swath gaps can be apparent in imagery.
The data are at the pixel resolution of the satellite. However, the locations and sizes of pixels depend on the orbit of the satellite and the angle between the sensor and the Earth.

To create Level 3 products, Level 2 data are averaged onto a spatial and temporal grid.

Usually, several pixels from the Level 2 data are averaged together in each grid, usually resulting in a coarser resolution product.

As part of this process, quality flags within Level 2 products are often used to screen out pixels before averaging. This excludes pixels where the retrieval algorithm indicates higher uncertainty, for example due to heavy cloud cover.
The same Level 2 product might be translated into multiple Level 3 products, depending on the desired spatial resolution, temporal resolution, or data quality. Common Level 3 temporal resolutions are daily or monthly averages.

Level 3 products result from aggregating Level 2 data onto a regular spatial and temporal grid.
The spatial and temporal consistency of Level 3 data make it ideal for analyses such as looking at changes over time and creating time averaged maps.

To produce Level 4 products, one or more Level 3 products may be combined with data from other sources.

Additional data sources can be atmospheric models, in-situ data, and/or demographic and health data.
Some applications require insights beyond what a single remote sensor can provide. By combining multiple observations and other data sources, we can generate integrated data products to address targeted information needs. Due to the additional processing required, Level 4 datasets are generally historical rather than near-real-time.

Level 4: Merged Satellite and Modeled Data

Level 4 products synthesize data from multiple sources to generate value-added products.
Some relevant air quality examples are surface level $PM_{2.5}$ or surface trace gas concentrations, or air quality exposure datasets.

One of the easiest ways to use remote sensing data when investigating air quality is to use true color and false color imagery to gain a qualitative understanding of the likely sources and impacts of aerosol air pollutants.

For certain major events, like fires and dust storms, dense plumes of aerosols can be seen in true color images, or identified in certain spectral bands via false color imagery.

True color imagery combines the red, green and blue visible wavelengths. We can detect aerosols with optical sensors as they block visible light. Most trace gases are not observable using true color imagery as they are not able to be detected with wavelengths in the visible portion of the electromagnetic spectrum.

Sun Glint

Over reflective surfaces, like snow and water, reflections of the sun show up as white, blurry patches, called sun glint, aligned with the satellite's orbital path. These features might also look like smoke, haze, dust, or clouds.

Smoke

Smoke often appears grey, brown, or black with the plume being smaller and denser near its source and spreading out and dissipating downwind.

Dust

Dust often appears tan or orange, and can be seen originating from bare soil or sand.

Clouds

Sometimes clouds can be hard to distinguish from smoke in true color images because they are often white and have different shapes depending on their altitude, wind speed, and direction.

Haze

Haze often appears grey or blue-grey over an area and usually occurs over cities.

When using true color imagery products, it is important to keep in mind that:

- It can be difficult to distinguish smoke from clouds or sun glint.

- Contextual information can help you interpret imagery, such as: fire locations, wind direction, and time of overpass for Low Earth Orbit (LEO) sensors.
- Geostationary Earth Orbit (GEO) sensors may have coarser spatial resolution than LEO sensors, making smaller smoke plumes or features more difficult to distinguish.
- They provide little to no observations at night.
- They show a top-down view of the whole atmosphere, not the surface-level air quality.
- They can be used to track plume transport over time.

Key Takeaway: Dense smoke is visible in true color imagery.

Geostationary sensors can provide more frequent observations, helping to visualize rapidly-evolving events like wildfires.

Key Takeaway: Frequent images from GEO sensors allow the pollutant plumes to be followed throughout the day.

False color imagery is created by assigning different color ranges to certain observed wavelengths. It is useful to enhance the visibility of aerosol distributions for applications such as daily air quality assessment and tracking pollution transport, wildfire smoke, and industrial emissions. False color imagery can also be used to identify areas that have been burned.

Instrument/Mission	Product/Algorithm	Key Characteristics/Recommendations	Visualization
VIIRS (Suomi-NPP*, NOAA-20, NOAA-21)	Corrected Reflectance (True Color)	True color imagery. Provides afternoon overpasses, about 1:30 PM local time. Available in Worldview starting 2015 (Suomi-NPP), 2018 (NOAA-20), and 2023 (NOAA-21).	Worldview-NOAA20-TrueColor (opens in a new tab)
MODIS** (Terra, Aqua)	Corrected Reflectance (True Color)	True color imagery. Terra: Provides morning overpasses, about 10 AM local time. Available starting 2000. Aqua: Provides afternoon overpasses, about 1:30 PM local time. Available starting 2002.	Worldview-Terra-TrueColor (opens in a new tab)
VIIRS (Suomi-NPP*, NOAA-20, NOAA-21)	Corrected Reflectance (Bands M3, I3, M11)	False color imagery. Useful for distinguishing between ice and clouds or vegetation versus bare surfaces to understand possible sources of interference with retrievals.	Worldview-NOAA20-FalseColor (opens in a new tab)
ABI (GOES-East, GOES-West)	GeoColor	True color imagery from a geostationary satellite instrument (GOES-East is the default for Aerosol Watch), offering 10-	AerosolWatch-GOESEast-GEOColor

		minute temporal resolution with coverage of North and South America.	(opens in a new tab)
ABI (GOES-East, GOES-West)	Dust RGB	False color imagery used to highlight blowing dust.	AerosolWatch-GOESEast-DustRGB

True color images can be used to visualize plumes of aerosol pollutants, like smoke or dust. The use of false color images can also take advantage of the spectral characteristics of different aerosols to make them more visually apparent. However, for more quantitative analysis, and to detect trace gas pollutants, we need to use higher-level data products.

Quantitative geophysical parameters relevant to air quality, such as AOD and trace gas columns, become available at data processing levels 2 and 3. Different retrieval algorithms applied to data from various remote sensors will yield distinct data products.

Example Applications: Identifying the presence of aerosols, mapping relative concentrations of trace gas pollutants over a region, tracking the long-range transport of pollutants

Considerations for Level 2 and 3 Products

- **Choose a product that meets your specific needs.** A satellite sensor can have more than one retrieval algorithm for the same geophysical parameter, resulting in slightly different data products for the same parameter.

- **Data products often have multiple versions.** New versions may incorporate algorithm improvements, bug fixes, or updated calibrations. Each version is often applied to the entire historical dataset through reprocessing to ensure consistency, and may have a staggered rollout depending on processing needs.
- **Always consult a particular dataset's documentation and metadata** for context on algorithmic considerations and product updates based on your research and application needs.

- **Satellite orbits determine data availability**. LEO instruments provide global coverage at fixed times daily, while GEO instruments capture diurnal patterns but only over specific regions.

Aerosol products use spectral information to distinguish between different aerosol types, such as dust, smoke, non-smoke fine particles (such as urban pollution), or background aerosols (such as sea salt).

AOD
Level 2 and 3 Aerosol Optical Depth (AOD) data products quantify aerosol presence in the atmospheric column.

Considerations When Using AOD Products
There are several different retrievals for AOD. These retrievals differ in several ways, including the wavelengths used, assumptions, and mathematical approaches. Because of this, different retrievals will perform better under certain conditions and are available at different temporal and spatial resolutions. Common AOD retrieval

algorithms include Dark Target (DT), Deep Blue (DB), and Multi-Angle Implementation of Atmospheric Correction (MAIAC). The best product choice will depend on your specific application. Common considerations for each retrieval algorithm include:

- **AOD represents all the atmospheric aerosol present.** It does not distinguish between different aerosol types and cannot identify surface-level-only aerosols.
- **Retrievals of AOD are difficult above clouds and areas of sun glint.**
- **Thick smoke can prevent reliable estimates of AOD.**
- **AOD is available only during the daytime.**

The AOD retrieval algorithm “Dark Target” makes certain assumptions about the reflectivity of land surfaces, and performs well over darker surfaces like forest and oceans. In the image, the red area shows that AOD is very high in the dense smoke plume southwest of Los Angeles. The yellow areas indicate lower AOD.

The missing pixels to the west of Los Angeles indicate the retrieval algorithm cannot determine AOD here due to the extremely dense smoke, which exceeds retrieval thresholds and may be misclassified as cloud cover.

There are also patches of missing data due to clouds, and over highly reflective surfaces like deserts. On the other hand, the retrieval detects higher AOD over the Pacific Ocean due to long range transport of wildfire smoke, as well as higher AOD due to a dust plume over the Gulf of California.

The AOD retrieval algorithm “Deep Blue” makes different assumptions about the surface reflectivity compared with “Dark Target”. It uses the high contrast between aerosols and land surfaces in the blue part of the spectrum to perform better over bright surfaces like deserts.
Compared to “Dark Target”, there is less missing data over the desert areas in California and Mexico.

However, it is facing similar challenges with clouds, snow, and the densest part of the smoke. Note also the differences at the northern end of the Gulf of California. “Dark Target” was providing a little more AOD data there, due to its better performance over water, so it caught more of the dust plume. This illustrates how different algorithms for the same geophysical parameter can produce slightly different results.

Key Takeaway: Missing data can be due to clouds, dense smoke, and reflective surfaces. Different AOD retrieval algorithms can give slightly different results.

Instrument/Mission	**Product/Algorithm**	**Key Characteristics/Recommendations**	**Visualization**
VIIRS (Suomi-NPP*, NOAA-20, NOAA-21)	Dark Target (opens in a new tab)	Continues the long-term AOD data record of MODIS with more modern instruments.	Worldview_AOD_ VIIRSDarkTarget_NOAA20 (opens in a new tab)

MODIS** (Terra/Aqua)	Merged DT/DB (opens in a new tab)	Combines outputs from both Dark Target and Deep Blue algorithms, choosing the most appropriate based on the land surface cover. This provides an easy-to-use product with enhanced spatial coverage.	Worldview_AOD_ _MODISMergedDTDB _Terra (opens in a new tab)
MODIS** (Terra/Aqua)	MAIAC (opens in a new tab)	High resolution, improved accuracy over heterogeneous areas such as cities. Can retrieve over thick aerosol layers.	Worldview_AOD_ MODISMAIAC_TerraAqua (opens in a new tab)

Angstrom Exponent
The Angstrom Exponent (AE) is a measure of how AOD varies with wavelength, providing information about aerosol particle size. AE can help distinguish between fine particles (like smoke or urban pollution) which have higher AE values, and coarse particles (like dust or sea salt) which have lower AE values.

Since AE depends on the accuracy of AOD retrievals, any uncertainties or limitations in the underlying AOD measurements will propagate through to the derived AE products.

The Angstrom Exponent derived from the Dark Target AOD illustrates some of the same limitations, including missing data over brighter surfaces. Even so, the distinction between the fine aerosols associated with smoke, which have high AE shown in purple and coarser aerosols associated with dust, which have low AE shown in green is clear. In conjunction with the AOD, this illustrates the relative presence and sizes of aerosol particles in different parts of the image.

Instrument/Mission	**Product/Algorithm**	**Key Characteristics/Recommendations**	**Visualization**
VIIRS (Suomi-NPP*, NOAA-20, NOAA-21)	Dark Target (opens in a new tab)	Continues data record of MODIS with more modern instruments. Ocean-only coverage. Note: Delivery of Suomi-NPP VIIRS observations will cease on November 1, 2026.	Worldview_AE_VIIRS (opens in a new tab) DarkTarget_NOAA20 (opens in a new tab)
VIIRS (Suomi-NPP*, NOAA-20, NOAA-21)	Deep Blue (opens in a new tab)	Continues data record of MODIS with more modern instruments. Land & ocean coverage. Note: Delivery of Suomi-NPP VIIRS observations will cease on November 1, 2026.	Worldview_AE_VIIRS _DeepBlue_NOAA20 (opens in a new tab)

MODIS** (Terra/Aqua)	Dark Target (opens in a new tab)	Terra: Data over oceans since 2000. Aqua: Data over oceans since 2002. Note: The Terra and Aqua missions are near the end of their life.	Worldview_AE_MODIS (opens in a new tab) DarkTarget_Terra (opens in a new tab)
MODIS** (Terra/Aqua)	Deep Blue (opens in a new tab)	Terra: Data over oceans since 2000. Aqua: Data over oceans since 2002. Note: The Terra and Aqua missions are near the end of their life.	Worldview_AE_MODIS DeepBlue_Terra

Aerosol Absorbing Index (AI)
The Aerosol Absorbing Index (AI, AAI, or UVAI) measures the absorption of ultraviolet (UV) radiation by aerosols. AI is useful for detecting and tracking absorbing aerosol plumes such as smoke from fires or mineral dust from desert regions.
Considerations for Aerosol Absorbing Index Products

- **AI products are typically found alongside ozone products** because they use the same wavelengths typical for ozone retrievals and they require ozone corrections.
- **UV wavelengths suffer less interference from clouds & thick smoke,** enhancing the spatial coverage of AI compared to other aerosol products.
- AI data products may have **reduced accuracy over snow/ice surfaces** due to high UV surface reflectance.
- **High values in AI data indicate either increased concentrations of absorbing aerosols or their presence at higher altitudes** where the UV signal is more easily detected by the satellite.

The UV Aerosol Absorption Index from Sentinel-5P TROPOMI shows a key difference from AOD and similar products by maintaining data coverage through clouds and the densest smoke areas, as seen in this January 9, 2025 image of the Los Angeles fires. Positive values of AAI, shown here in red, indicate the presence of many absorbing aerosols, which in this case is smoke. The cyan, green, and yellow colors at the north end of the Gulf of California indicate the presence of dust there. Snow-covered areas remain the primary limitation for data availability, with low values regardless of aerosol presence.

Key Takeaway: AI is an indicator of aerosol composition (e.g., absorbing smoke or dust vs. less absorbing urban pollution or volcanic emissions), and is less susceptible to missing data due to clouds or dense smoke.

Getting Started with Aerosol Absorption Index Products
This table provides some additional easily accessible and commonly used Level 2 and 3 Aerosol Absorption Index products.

Instrument/Mission	Product/Algorithm	Key Characteristics/Recommendations	Visualization
OMI (Aura)	OMAERUV (opens in a new tab)	Provides global coverage since 2004.	Worldview - OMAERUV (opens in a new tab)
OMPS (Suomi-NPP)	OMPS-NPP L2 NM (opens in a new tab)	Continues the global coverage record of OMI with a more modern instrument (since 2012).	Worldview - OMPS (opens in a new tab)
TEMPO (Intelsat-40e)	TEMPO_O3TOT_L3_V04 (opens in a new tab)	Provides hourly coverage for North America from a GEO instrument. Level 3 data aggregates scan data into hourly images, which are usually simpler to work with.	Worldview - TEMPO O3

Getting Started with Aerosol Absorption Index Products

This table provides some additional easily accessible and commonly used Level 2 and 3 Aerosol Absorption Index products.

Instrument/Mission	Product/Algorithm	Key Characteristics/Recommendations	Visualization
OMPS (Suomi-NPP, NOAA-20, NOAA-21)	OMPS_N20_ (opens in a new tab) NMUVAI_L2_NRT (opens in a new tab)	Provides global coverage since 2012. Note: Delivery of Suomi-NPP VIIRS observations will cease on November 1, 2026.	Worldview_AI _OMPS_NOAA20 (opens in a new tab)
TROPOMI (Sentinel-5p)	L2__AER_AI (opens in a new tab)	Provides global coverage since 2018.	JStarMapper_AI (opens in a new tab) _TROPOMI

			(opens in a new tab) (Choose Aerosol Index from the dropdown menu)
OMI (Aura)	OMAERUV (opens in a new tab)	Provides global coverage since 2004. Note: The Aura mission is near the end of its life.	Worldview_AI_OMI (opens in a new tab)
TEMPO (Intelsat-40e)	TEMPO_O3TOT_L3_V03 (opens in a new tab)	Provides hourly coverage for North America from a GEO instrument. Level 3 data aggregates scan data into hourly images, which are usually simpler to work with.	Worldview_AI_TEMPO_L3 (opens in a new tab)

Aerosol Type
Aerosol type products classify atmospheric aerosols according to their source and composition, providing essential information for air quality. Advanced retrieval algorithms use spectral information to distinguish between different aerosol types, such as dust, smoke, non-smoke fine particles (such as urban pollution), or background aerosols (such as sea salt).
Considerations for Aerosol Type Products

- Aerosol type classification **generally works best with moderate-to-high aerosol loading,** because the classification needs sufficient aerosol signal strength to distinguish different types from surface effects.
- **Some aerosol type products are based on regional information only** (e.g., North American data), but used globally, therefore aerosol type accuracy may be lower in regions with different surface conditions or climates.

While particle types can be inferred using parameters like the Ångström Exponent, some data products go further, classifying aerosol types by their spectral signatures. This image shows Deep Blue retrieval classification: red indicates smoke, brown is dust, and green denotes "fine-dominated" aerosols, which are mostly small particles, here likely smoke, urban haze, or fine particle pollution. Grey signifies "background" aerosols—concentrations are too low for specific type categorization; over oceans and coasts, these may be sea salt.

Key Takeaway: Aerosol type products assign aerosols into categories based on spectral signatures. This is easy to interpret, but looking at AOD and AE separately can also provide context when a type might be in doubt.

Getting Started with Aerosol Type Products
This table provides some additional easily accessible and commonly used Level 2 and 3 Aerosol Type products.

Instrument/Mission	Product/Algorithm	Key Characteristics/Recommendations	Visualization
VIIRS (Suomi-NPP*, NOAA-20, NOAA-21)	Deep Blue (opens in a new tab)	Global coverage since 2012. Note: Delivery of Suomi-NPP VIIRS observations will cease on November 1, 2026.	Worldview_AerType_ (opens in a new tab) VIIRSDeepBlue_NOAA20 (opens in a new tab)
ABI (GOES-East, GOES-West)	Smoke Dust Mask (opens in a new tab)	Geostationary instruments provide 10-minute temporal resolution data over much of North and South America. Algorithm identifies plumes of smoke or dust aerosols.	AerosolWatch_ (opens in a new tab) SmkDstMsk_GOESEast (opens in a new tab) (Select 'Smoke Dust Mask' from the dropdown menu.)

Aerosol Products Summary
Aerosol products like AOD, Angstrom Exponent, and Aerosol Absorbing Index (AI) provide information about the abundance, size, and absorption properties of atmospheric aerosols. Used together, these products can provide information about the composition and source of aerosols. Aerosol Type data products use this combined information to classify aerosols into categories (e.g., smoke, dust).
Aerosol data products frequently contain data gaps across various locations and time periods, primarily caused by cloud cover, highly reflective surfaces, and extremely dense smoke plumes.
Looking at long-term averages using gridded Level 3 products can help overcome these missing data, but when missing data is caused by systematic factors such as seasonal cloud cover, this can skew the analysis towards over-representing conditions which allow aerosol retrievals to function well.

Remote Sensing of trace gases provides valuable information about the sources, transport, and chemical evolution of these pollutants.

Nitrogen Dioxide
Nitrogen dioxide (NO_2) is a trace gas with a short lifetime. Tropospheric column NO_2 measurements represent the column rather than near-surface concentrations where human exposure occurs, but can help identify pollution sources and monitor air quality.

Considerations for NO_2 Products

- NO_2 products usually **specify whether the data is for the total column, stratospheric column, or tropospheric column.**

- Tropospheric column NO_2 retrievals provide relevant information for air quality because the majority of NO_2 emissions originate at the Earth's surface, and the **relatively short lifetime of NO_2** (a few hours to a day) means that NO_2 is **often found close to its emission source.**
- **GEO satellites offer sub-daily NO_2 observations for daytime only.** For example, TEMPO provides hourly measurements of NO_2 over North America, which is very useful for tracking diurnal variations or weekday/weekend effects in NO_2 pollution.

Coverage of TEMPO hyperspectral, geostationary Level 2 data moves from East to West as the instrument scans over the course of an hour, arriving over the LA wildfires at the end of a scan cycle. As we scroll through time, we see the coverage moving from East to West as the instrument scans over the course of an hour, arriving over the LA wildfires at the end of a scan cycle.

The Level 3 TEMPO product combines data from the entire hourly scan, regridded to a consistent spatial resolution. High values of NO2 associated with the two wildfire events are clearly visible. The NO2 is concentrated much closer to the sources, whereas the smoke in the aerosol products was spread further downwind. Plumes associated with urban areas like Bakersfield, San Diego, and Tijuana are also visible here.

Key Takeaway: Level 2 data are usually associated with an individual orbit swath (LEO) or scan (GEO). Level 3 data generally aggregate multiple orbits or parts of scans into a single data product, with a specified spatial and temporal resolution.

Getting Started with NO_2 Products

This table provides some additional easily accessible and commonly used tropospheric column Level 2 and 3 NO_2 products.

Instrument/Mission	Product/Algorithm	Key Characteristics/Recommendations	Visualization
TROPOMI (Sentinel-5P)	S5P_L2__NO2____HiR (opens in a new tab)	Provides global coverage since 2018. Higher spatial resolution and better coverage compared to OMI.	Worldview_NO2_TROPOMI (opens in a new tab) CopernicusBrowser_NO2_TROPOMI (opens in a new tab)
OMI (Aura)	OMNO2d (opens in a new tab)	Provides global coverage since 2004. Note: The Aura mission is near the end of its life.	Worldview_NO2_OMI (opens in a new tab)

TEMPO (Intelsat-40e)	TEMPO_NO2_L3_V04 (opens in a new tab)	Provides hourly coverage for North America from a GEO instrument. Level 3 data aggregates scan data into hourly images, which are usually simpler to work with.	Worldview_NO2_TEMPO

Sulfur Dioxide

Sulfur dioxide (SO_2) is a trace gas emitted from both anthropogenic sources like coal-fired power plants and natural sources such as volcanic eruptions. SO_2 satellite measurements are particularly useful for tracking large emission sources and volcanic plumes.

Considerations for SO_2 Products

- Tropospheric SO2 retrieval algorithms require assumptions about the emission height, such as the altitude of volcanic eruption plumes. Because of this, products may indicate a specific layer of the atmosphere (e.g., planetary boundary layer, lower/middle/upper troposphere, or stratosphere).

Getting Started with SO_2 Products

This table provides some additional easily accessible and commonly used Level 2 and 3 total column SO_2 products.

Instrument/Mission	Product/Algorithm	Key Characteristics/Recommendations	Visualization
TROPOMI (Sentinel-5P)	S5P_L2__SO2____HiR (opens in a new tab)	Provides total column SO_2 with global coverage since 2018. Higher spatial resolution and better coverage compared to OMI.	Worldview_SO2_TROPOMI (opens in a new tab)
OMPS (Suomi-NPP, NOAA-20, NOAA-21)	NMSO2_PCA (opens in a new tab)	Provides total column SO_2 with global coverage since 2012.*	Worldview_SO2 (opens in a new tab) _PBL_OMPS_NOAA20 (opens in a new tab)
OMI (Aura)	OMSO2e (opens in a new tab)	Provides total column SO_2 with global coverage since 2024.* This level 3 product is available daily on a 0.25 degree latitude-longitude grid.	Worldview_SO2_PBL_OMI

Carbon Monoxide

Carbon monoxide (CO) is a trace gas produced by incomplete combustion and has a relatively long atmospheric lifetime, making it useful as a tracer for pollution transport. CO satellite measurements can track the movement of pollution plumes over large distances.

Considerations CO Products

- **CO products are usually provided as total or partial column** or at specific atmospheric pressure levels**.**
- **CO is observable at night using infrared radiation** re-emitted from the Earth.

The MOPITT (Measurement of Pollution in the Troposphere) instrument aboard the Terra satellite has retrieval algorithms implemented for Carbon Monoxide concentrations in the atmosphere. The spectral signature of CO includes the IR range and often allows these algorithms to function even at night. This enables data products like MOPITT CO to include both daytime and nighttime information.

Key Takeaway: CO retrievals that use IR wavelengths can provide observations for both day and night.

Getting Started with CO Products

This table provides some additional easily accessible and commonly used Level 2 and 3 CO products.

Instrument/Mission	**Product/Algorithm**	**Key Characteristics/Recommendations**	**Visualization**
TROPOMI (Sentinel-5P)	S5P_L2__CO_____HiR (opens in a new tab)	Provides total column CO with global coverage since 2018.	Copernicus Browser _CO_TROPOMI (opens in a new tab)
MOPITT (Terra)	MOP03J (opens in a new tab)	MOPITT provides total-column CO, which is useful for tracing long-range pollutant transport. The retrieval also estimates the near-surface CO, which is more directly useful for air quality.	Worldview_CO _MOPITT (opens in a new tab)
AIRS (Aqua)	AIRS2RET (opens in a new tab)	AIRS provides profiles of CO at different altitudes; 500 hPa represents the pressure at about 5.5 km altitude, where weather patterns can transport CO far from it sources, so examining CO at this altitude can help trace long-range transport of air pollution.	Worldview_CO (opens in a new tab) _AIRS

Trace Gas Products Summary
Level 2 and 3 trace gas and aerosol products can tell us about the presence and types of different pollutants in the atmosphere. This allows us to make inferences about the likely sources of these pollutants, where they are being transported to, and the typical spatial and temporal patterns of their variability.
However, these products do not directly provide surface level concentrations of these pollutants, which is the information we need to determine air quality. For this, we need to take things a step further.

Satellite data alone cannot provide us information about air quality at the surface. As we've seen, the level 2 or 3 satellite products provide information about the atmospheric column, not near-surface pollutant concentrations that are needed to evaluate air quality. In such cases, we need to bring in data from other sources and combine it with satellite information to derive the information we want. This is the process for developing Level 4 data products.
Common Applications: Mapping surface concentrations of pollutants in areas without in-situ monitors, studying the exposure of populations to air pollutants, forecasting future air quality.

Though there can be many different kinds of Level 4 data products, we will focus on three types that are most commonly produced by NASA or NASA-funded research for air quality applications:

-Reanalysis
A model run with additional observation data to simulate past conditions.

-Forecast
A model that simulates future conditions.

-Hybrid or Fusion
Information is combined from multiple sources, including models, reanalyses, and other observation data.

Reanalysis Products
A **Reanalysis** is a model which is run in a retrospective mode to simulate past conditions of the Earth's systems, and in the case of air quality we are looking at the atmosphere and the air pollutants within it. In a reanalysis, data assimilation is used to incorporate observations from satellites and other sources, bringing the model's state into better alignment with the observed reality.

Data Assimilation: The process of incorporating observations with a model output to produce the best estimate of the atmosphere, land, and/or ocean conditions.

Reanalyses of the Earth's atmosphere, which include simulations of aerosols and trace gases, can provide surface-level pollutant concentration information. This information does not come from the satellite only, but is from a model informed by satellite data. For example, reanalyses providing $PM_{2.5}$ information commonly

assimilate satellite AOD to provide information about the total amount of aerosols in the atmosphere, and the model determines how much of the aerosol is near the surface where it can impact air quality.

Considerations for Reanalysis Products

- Provide near-surface concentrations relevant to assessing air quality
- Can fill gaps in satellite data
- Reanalysis spatial resolution is limited to the resolution of the model used, which is often coarser than satellite data.
- Reanalysis products often run days or weeks behind real-time, allowing robust quality control for the assimilated observations.
- Long-running reanalyses may span multiple satellite missions, or even go back further than satellite data were available, so there can be "jumps" or discontinuities in the reanalysis as new observations are introduced into the data assimilation system.
- Certain geophysical parameters may be simulated by the reanalysis, but not observed with any of the assimilated data; these parameters are "unconstrained", defaulting to the model calculations.

Key Takeaway: Integrating information from multiple data sources into atmospheric models can help us understand the processes impacting air quality and fill gaps in satellite data.

Getting Started with Reanalysis Products

Here are a few commonly used Level 4 Reanalysis products relevant to air quality you might want to use to get started.

Organization	Product/Algorithm	Key Characteristics/Recommendations	Visualization
NASA Global Modeling & Assimilation Office (GMAO) (opens in a new tab)	MERRA-2 (opens in a new tab)	Retrospective global reanalysis providing data since 1980, running about a month behind present time. Includes outputs relevant to air quality (Aerosol types, $PM_{2.5}$, CO, SO_2, Total Column O_3). Multiple satellite, ground-based, and airborne sensors contribute inputs to the model.	FLUID_PM25 (opens in a new tab) _MERRA2 (opens in a new tab)
European Centre for Medium-range Weather Forecasts (ECMWF) (opens in a new tab)	EAC4 (opens in a new tab)	Covers a period from 2003 to about one year behind the present time. Includes several aerosols and trace gases relevant to air quality ($PM_{2.5}$, PM_{10}, NO_2, O_3, SO_2, CO). Multiple satellite, ground-based, and airborne sensors contribute inputs to the model.	Copernicus Data Service (opens in a new tab) (download data)

Forecast Products

A forecast is a model which is run forward from the present time to simulate future conditions of the Earth's systems (e.g., future air quality). Like a reanalysis, data assimilation may be used to establish the forecast's initial conditions.

Initial conditions are the starting state of the model, from which it is run forward to produce the forecast. Generally, the more true-to-reality the initial conditions are, the better the forecast will be.

Forecasts are our way of predicting what the future will bring for air quality. While satellite data are available to inform the forecast's initial conditions, we cannot have observations of the future, so we rely on the model to tell us how these conditions will change in the near future.

Considerations for Forecast Products

- They provide future near-surface concentrations relevant to assessing air quality.
- Forecast spatial resolution is limited to the resolution of the model used, which is often coarser than satellite data.
- If satellite data used to establish forecast initial conditions are delayed, unavailable, or have errors which could not be corrected in time for the forecast, the forecast accuracy will be reduced.
- Forecasts take some time to run; by the time the outputs become available, it may already be a few hours into the forecast period.
- Because no new observations are available to constrain them, today's air quality forecasts are only useful out to about 10 days or less.
- Because the model cannot predict certain extreme events (e.g., a wildfire or volcanic eruption), the air quality impacts of that event will not show up in the forecast; similarly, a model cannot predict when a fire will be put out, and may show it impacting air quality long after it actually ends.

The Goddard Earth Observing System Composition Forecast, GEOS-CF — assimilates OMI satellite ozone data to simulate atmospheric composition, including air-quality-relevant near-surface ozone.

While satellites cannot directly observe near-surface ozone, their data combined with models like GEOS-CF enable usable surface-level air quality information

Key Takeaway: Forecasts can predict how air quality will change in the near future, given information on present conditions. Forecasts will only capture the impacts of the emissions sources (e.g., fires) they know about; they cannot predict whether a new emission source will appear.

Getting Started with Forecast Products

Here are a few commonly used Level 4 Forecast products relevant to air quality you might want to use to get started.

Organization	Product/Algorithm	Key Characteristics/Recommendations	Visualization
NASA Global Modeling & Assimilation Office (opens in a new tab)	GEOS-Forward Processing (opens in a new tab) (GEOS-FP)	Provides global hourly forecasts of meteorology, aerosols, and CO concentrations for the next 10 days. Data from many different satellites are assimilated, along with global meteorological data from many sources.	FLUID_PM25 (opens in a new tab) _GEOSFP (opens in a new tab)

	(opens in a new tab)		
NASA Global Modeling & Assimilation Office (opens in a new tab)	GEOS Composition Forecast (opens in a new tab) (GEOS-CF) (opens in a new tab)	Provides global hourly forecasts of atmospheric composition for the next 5 days. Includes outputs relevant to air quality ($PM_{2.5}$, PM_{10}, NO_2, O_3, SO_2, CO). Ozone data from Aura OMI and MLS instruments are assimilated.	FLUID_GEOSCF; GEOS-CF Interactive Map (opens in a new tab)
European Centre for Medium-range Weather Forecasts (ECMWF) (opens in a new tab)	CAMS (opens in a new tab)	Provides global hourly forecasts of atmospheric composition for the next 5 days. Includes outputs relevant to air quality ($PM_{2.5}$, PM_{10}, NO_2, O_3, SO_2, CO). Data from many different satellites are assimilated, along with global meteorological data from many sources.	CAMS Charts

Hybrid and Fusion Products

Hybrid or fusion datasets combine information from multiple sources, including models, reanalyses, satellite data, and/or ground-based monitors. They may also incorporate external data relevant to air quality, like population density or emission sources. Fusion or hybrid products may use a variety of different techniques, like regression or machine learning, to integrate these data.

A hybrid or fusion product might be created to:

1. Incorporate additional data which were not included in a reanalysis, without performing a new reanalysis.
2. Generate an output at a higher spatial resolution than a reanalysis or forecast is able to achieve.
3. Correct for errors or biases in other products to make them better suited to a particular region or application.

Considerations for Hybrid and Fusion Products

There are many reasons why and how hybrid or fusion products are created. These considerations cover some broad commonalities of hybrid and fusion products for estimating near-surface pollutant concentrations, but may not apply to every product.

- Can combine the strengths of different sources while compensating for weaknesses.
- Can provide finer spatial resolution than model or satellite data by integrating high-resolution information on related variables (e.g., traffic density).
- It may take months or years to collect and process all the data needed to generate these products.
- Products calibrated to work well in specific regions may struggle in others; particularly, regions with few in-situ air quality monitors lack data to calibrate new surface concentration products or evaluate existing ones.

Key Takeaway: Hybrid or fusion products bring in additional data sources to further refine estimates. These products can take years to develop, which is why this example is from 2024, a year before the fire.

Getting Started with Hybrid and Fusion Products

Here are a few commonly used Hybrid and Fusion Level 4 products relevant to air quality you might want to use to get started.

Organization	Product/Algorithm	Key Characteristics/Recommendations	Visualization
Washington University St. Louis Atmospheric Composition Analysis Group (opens in a new tab)	Global Annual Gridded PM2.5, V5.GL.04 (opens in a new tab)	Estimates of global annual-average $PM_{2.5}$ concentrations from several input datasets, including AOD from MODIS and other instruments. Multi-year averages (2000-2010, 2010-2012) are available for visualization in Worldview; full data (1998-2022) can be downloaded from Earthdata.	Worldview_sfcPM25 (view data), (opens in a new tab) Earthdata_sfcP_M25 (Download Data) (opens in a new tab)
NASA Health and Air Quality Applied Sciences Team (opens in a new tab)	MERRA2_CNN _HAQAST_PM25 (opens in a new tab)	While the MERRA-2 reanalysis includes global $PM_{2.5}$ estimates, researchers have noted its biases compared to ground-based measurements. This dataset was created using a machine learning technique to detect and correct for these biases, using additional ground-based $PM_{2.5}$ measurements not assimilated by MERRA-2.	Giovanni_sfcPM25 (opens in a new tab) _HAQAST (opens in a new tab) (Analyze Data Online)

The previous reanalysis, forecast, and hybrid examples focused on estimating near-surface pollutant concentrations relevant to air quality. Other Level 4 products are created for different purposes, such as estimating the amounts of pollutants emitted by different sources or assessing the health impacts of pollutant exposure.